SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:
Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL6)

Assessment Tool Weightage: 30%

QUESTIONS: 5 **Total Marks:25**

Annexure A

Comparative Analysis

Code of Conduct:

I **N. HEMA SREE (192511100)** *certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **N. HEMA SREE**

Annexure A

Comparative Analysis

1. Star topology is generally more expensive than bus topology because it requires a central device such as a switch or hub and more cabling to connect each node individually. In contrast, bus topology uses a single backbone cable, making it low-cost and easier to install. However, in terms of reliability, star topology is superior because failure of one node does not affect others, whereas a failure in the main bus cable can bring down the entire network. Performance is also better in star topology since data transmission is managed through a central device, reducing collisions, while bus topology suffers from performance degradation as more devices are added due to increased data traffic on the shared medium.

ANSWER:
Star topology is generally more expensive than bus topology because it requires a central networking device such as a switch or hub and separate cables for connecting each device. In contrast, bus topology uses a single backbone cable, making it cheaper and easier to install, especially for small networks.
In terms of **reliability**, star topology is superior because the failure of one connected device or cable affects only that particular device. However, if the central switch or hub fails, the entire network can be disrupted. In bus topology, a failure or break in the main backbone cable can affect the communication of all connected devices.
Regarding **performance**, star topology provides better performance because the switch manages communication between devices and reduces unnecessary traffic and collisions. Bus topology uses a shared communication medium, so as more devices are added, network traffic increases and performance decreases.
Example: Modern offices and computer laboratories commonly use star topology because they require better performance and easier fault management, whereas bus topology may be suitable for a small and temporary network.
Conclusion: Although bus topology is cheaper, star topology is a better choice for most modern networks because it provides better performance, reliability, scalability, and easier maintenance.

2. Mesh topology offers very high fault tolerance because every node is connected to multiple other nodes, providing multiple paths for data transmission. If one link fails, data can still be routed through alternate paths without disrupting communication. In contrast, star topology has moderate fault tolerance; while failure of individual nodes does not affect the network, failure of the central hub or switch can cause the entire network to stop functioning. Therefore, mesh topology is more robust and reliable in critical environments compared to star topology.

ANSWER:
Mesh topology provides **very high fault tolerance** because multiple communication paths exist between nodes. In a full mesh network, every node is directly connected to every other node. Therefore, if one connection fails, data can travel through another available path, allowing communication to continue.
Star topology provides **moderate fault tolerance**. If a single node or its connecting cable fails, the remaining devices continue to operate normally. However, the central switch or hub represents a single point of failure. If it fails, communication between the connected devices is interrupted.

The difference becomes especially important in critical environments where continuous communication is required. Mesh topology offers greater redundancy and reliability, but this comes at the cost of additional cables, ports, installation effort, and maintenance.
Example: Mesh topology is useful in critical communication systems where alternative paths are important, while star topology is commonly used in offices because it provides a practical balance between reliability and cost.
Conclusion: Mesh topology is more fault tolerant and reliable than star topology because it provides multiple alternative communication paths. However, star topology is more practical for normal organizational networks because of its lower complexity and cost.

3. Bus topology is simple and easy to install because it uses a single backbone cable with minimal connections, making it suitable for small networks. On the other hand, mesh topology is highly complex to install as each node must be connected to every other node, requiring a large number of cables and ports. This complexity increases significantly as the number of devices grows, making mesh topology time-consuming and costly to implement compared to the straightforward setup of bus topology.

ANSWER:
Bus topology is relatively simple to install because all devices are connected to a single backbone cable. It requires fewer cables and networking components, making the installation process straightforward and inexpensive. However, proper termination of the backbone is required, and locating faults in the main cable can be difficult.
Mesh topology is considerably more complex to install because multiple connections are required between nodes. In a **full mesh topology**, every node is directly connected to every other node. As the number of devices increases, the number of connections increases rapidly. For n devices, a full mesh requires $\mathbf{n(n-1)/2}$ **links**, which makes large networks expensive and difficult to install.
For example, a network containing 5 devices requires 10 direct links in a full mesh, whereas a bus network can connect the same devices using a single backbone cable.
Conclusion: Bus topology is much simpler, cheaper, and faster to install, making it suitable for small or temporary networks. Mesh topology provides better redundancy but is more complex and costly to implement.

4. Hybrid topology is highly scalable because it combines two or more different topologies (such as star, mesh, or bus), allowing the network to expand flexibly according to organizational needs. New segments can be added without affecting the existing network structure. In comparison, star topology is also scalable but to a limited extent, as it depends on the capacity of the central device (switch or hub). Once the central device reaches its port limit, expansion requires additional hardware. Thus, hybrid topology provides greater scalability than star topology.

ANSWER:
Hybrid topology provides greater **scalability and flexibility** because it combines two or more different topologies, such as star, bus, or mesh. Different sections of an organization can use different network structures according to their requirements. New segments can be added without completely redesigning the existing network.
Star topology is also scalable, but its expansion depends mainly on the capacity of the central switch or hub. If all available ports are occupied, an additional switch or other networking equipment is required to connect more devices.
Hybrid topology is particularly useful for large organizations with multiple departments or buildings. For example, an organization may use star topology within individual departments and connect these departments using another topology or backbone structure.

<u>Conclusion:</u> Hybrid topology provides greater scalability and flexibility than a basic star topology because different network segments can be expanded or modified according to organizational requirements. However, the increased flexibility also results in greater design and management complexity.

5. In terms of maintenance, star topology is relatively easy to manage because issues can be quickly identified and isolated to a specific node or cable. Mesh topology, while highly reliable, is difficult to maintain due to its complex structure and large number of connections, making troubleshooting time-consuming. Bus topology is harder to maintain than star because faults in the backbone cable can disrupt the entire network and are difficult to locate. Hybrid topology has moderate maintenance complexity; while it benefits from flexibility, managing multiple combined topologies requires careful monitoring and skilled administration.

ANSWER:

Star topology is generally the easiest to maintain because each device has an individual connection to the central switch or hub. If a particular computer or cable fails, administrators can identify and isolate the affected connection without disrupting the rest of the network.

Mesh topology is highly reliable but more difficult to maintain because it contains many connections. A fault can be isolated through alternative paths, but the large number of links makes inspection, troubleshooting, and maintenance more complex.

Bus topology is relatively simple in structure but can be difficult to troubleshoot. A fault in the backbone cable or termination can affect communication across the network, and locating the exact fault can take considerable time.

Hybrid topology has **moderate to high maintenance complexity** because it combines different topologies. Administrators must understand and manage multiple network structures, devices, and communication paths. However, its flexibility allows faults to be isolated within particular network segments.

<u>Example:</u> In a college or corporate network, star topology is often preferred because technicians can easily identify problems affecting individual computers or cables.

<u>Conclusion:</u> Star topology is generally the easiest to maintain, while mesh topology provides the highest redundancy but requires more complex maintenance. Bus topology is simple but can be difficult to troubleshoot, and hybrid topology provides flexibility at the cost of increased management complexity.